

An army outside a city

- Imagine three army camps surrounding a fortified city.
- Their generals must choose: ATTACK together or RETREAT together.
- They communicate by sending messengers between camps.

This is a thought experiment about coordination, not a documented battle.

What if one general is a traitor?

- A loyal general follows the agreed procedure.
- A traitor may lie, stay silent, or tell different camps different stories.
- The loyal generals do not know who the traitor is.

The procedure must work even when the traitor tries to make it fail.

Our three generals

General	Role
C	Commander: sends the proposed order
Alice	Lieutenant: receives and checks the order
Bob	Lieutenant: receives and checks the order

- Any one of these three may be the traitor.
- For now, messages carry no verifiable signature.

We follow Alice and Bob as they decide what order to obey.

First, a day with no betrayal

Message	Content
C to Alice	ATTACK
C to Bob	ATTACK
Alice to Bob	C told me ATTACK
Bob to Alice	C told me ATTACK

Alice and Bob both choose ATTACK. Nothing conflicts.

Now Bob lies

What happens	Message
Loyal C tells Alice	ATTACK
Loyal C tells Bob	ATTACK
Traitor Bob tells Alice	C told me RETREAT

Alice hears two conflicting claims. In this situation she should follow loyal C.

Now the commander lies

What happens	Message
Traitor C tells Alice	ATTACK
Traitor C tells Bob	RETREAT
Loyal Bob tells Alice	C told me RETREAT

Alice receives exactly the same initial messages as on the previous slide.

Alice cannot see behind the messages

Alice receives	Bob lies	C lies
Direct order from C	ATTACK	ATTACK
Bob's report	RETREAT	RETREAT

Different hidden situations can look identical to one loyal participant.

What must a successful procedure guarantee?

- **Agreement:** loyal Alice and loyal Bob choose the same order.
- **Validity:** if C is loyal, loyal lieutenants follow C's order.
- **Termination:** loyal lieutenants eventually decide.

Always choosing RETREAT gives agreement, but can violate validity.

Why the obvious fixes fail

Proposed rule	A situation that defeats it
Always trust C	C sends conflicting orders
Always trust the other lieutenant	That lieutenant lies
Retreat whenever reports conflict	Loyal C says ATTACK; a lieutenant lies
Wait for everyone to confess	A traitor never confesses

A rule must work for every permitted location and behavior of the traitor.

The contradiction, in ordinary words

- If loyal C says ATTACK while Bob lies, Alice must choose ATTACK.
- A lying C can create the same apparent situation for Alice.
- Symmetrically, Bob may be unable to distinguish a loyal RETREAT order from a lying C.
- With C lying, loyal Alice can be forced toward ATTACK and loyal Bob toward RETREAT.

Indistinguishability means a participant cannot tell two possible executions apart.

The three-general impossibility result

- Three generals cannot guarantee all the requirements against one traitor in the oral-message model.
- Receivers know the immediate sender. Messages have no transferable signatures.
- Loyal-to-loyal delivery is reliable, and round deadlines reveal omissions.
- The result rules out a guaranteed solution, even with more discussion.

The impossibility belongs to these assumptions. Changing them can change the answer.

The general threshold

$$n \geq 3f + 1$$

Possible traitors f	Total participants needed
1	4
2	7
3	10

Here n counts everyone, including the commander. This is the classical oral-message setting.

A fourth general: Carol

- Now we have C, Alice, Bob, and Carol, with at most one traitor.
- Round 1: C sends an order to each lieutenant.
- Round 2: each lieutenant tells the other two what C sent.
- Each lieutenant takes the majority of three entries: its direct order and two reports.

A missing or malformed report becomes the agreed default RETREAT.

Four generals: Carol lies

Loyal recipient	Own order	Other loyal report	Carol's report	Decision
Alice	ATTACK	ATTACK	RETREAT	ATTACK
Bob	ATTACK	ATTACK	RETREAT	ATTACK

With loyal C, two matching honest entries outweigh Carol's one false entry.

Four generals: C lies

Loyal recipient	Alice position	Bob position	Carol position	Decision
Alice	ATTACK	RETREAT	ATTACK	ATTACK
Bob	ATTACK	RETREAT	ATTACK	ATTACK
Carol	ATTACK	RETREAT	ATTACK	ATTACK

If C is the only traitor, all three lieutenants relay honestly and reconstruct the same vector.

LSP: the algorithm behind our four generals

- LSP names Lamport, Shostak, and Pease.
- Their oral-message algorithm is written $OM(m)$.
- The parameter m is the number of traitors it is designed to tolerate.
- Our two-round four-general example is $OM(1)$.

$OM(0)$ means direct delivery; $OM(1)$ adds one level of cross-checking.

LSP OM(m), step by step

- 1. The commander sends its value to every lieutenant.
- 2. Lieutenant i stores the received value as v_i ; silence becomes RETREAT.
- 3. If $m > 0$, each lieutenant distributes v_i to the other lieutenants using OM($m - 1$).
- 4. Each lieutenant takes a majority over its own entry and the recursively obtained entries.

Each subproblem excludes its commander from further recursive recipient roles.

LSP OM(2): a small trace inside seven generals

Round	A representative message path
1	C tells Alice ATTACK
2	Alice tells Bob: C told me ATTACK
3	Bob tells Carol: Alice reported ATTACK

- With up to two traitors, use seven generals in total.
- Every eligible path runs, not just the representative path above.
- Combine reports from the deepest level upward.

Three rounds provide two levels of checking beyond the original command.

What LSP teaches, and what it costs

Feature	LSP oral-message algorithm
Fault bound	Classical nontrivial case: n at least $3m+1$
Rounds	$m+1$ synchronous rounds
Traffic	Recursive forwarding along many distinct paths
Purpose	Establish that agreement is possible under the model

The four-general protocol is an explanation of possibility, not a ready-made blockchain system.

Sealed orders: signatures

- Imagine an order with a seal that others can verify but cannot forge.
- Alice can forward C's sealed ATTACK order to Bob.
- Bob can check C's endorsement without trusting Alice's description.
- In computers, a digital signature provides this kind of evidence.

A signature proves endorsement of specific content. It does not prove the content is true.

Three generals with signed orders

Step	What loyal Alice and Bob learn
C signs conflicting orders	Alice gets ATTACK; Bob gets RETREAT
They forward the signed orders	Both see the same two signed values
They use a common rule	One value: use it. Conflict or none: RETREAT

With a loyal C, no traitor can manufacture a different C-signed order.

The same problem inside a computer network

Army story	Distributed system
General	Server, replica, or validator
Order	Proposed operation or block
Messenger	Network message
Traitor	Byzantine process
Common action	Consistent decision

Several computers must keep a common record despite some arbitrary failures.

What Byzantine fault tolerance means

- A Byzantine fault is arbitrary deviation from the protocol.
- It can come from a malicious compromise or a severe implementation fault.
- A BFT protocol preserves stated guarantees despite up to its allowed fault bound.
- It need not identify or repair every faulty participant.

Tolerance means honest participants remain protected even while faulty ones behave badly.

Safety, validity, and liveness

Property	Plain meaning	Ledger example
Safety	No conflicting final decisions	Honest replicas do not finalize conflicting blocks
Validity	Decisions satisfy the stated rule	A decided operation meets the chosen validity rule
Liveness	Honest work eventually progresses	A suitable proposal can eventually commit

A network disruption may stop progress while the protocol still protects safety.

What a Byzantine adversary controls

Allowed capability	Important boundary
Control up to f participants	Honest participants still follow the protocol
Coordinate controlled participants	Several faults may act as one attacker
Send conflicting messages or stay silent	The attacker controls its own sends
Use corrupted participants' keys	It cannot forge uncompromised signatures

Corrupting a node and controlling network delivery are separate powers.

What the network allows

Network model	Delivery guarantee	Meaning of a timeout
Synchronous	Known upper delay bound	A deadline can establish a missing message
Asynchronous	No known delay bound	Slow and faulty cannot be distinguished by time
Partially synchronous	A bound holds after an unknown time	Early timeouts may be wrong

A delayed honest message can look like silence. The timing model determines what you may infer.

One explicit adversarial model for BFT

- **Membership:** $n = 3f + 1$ known participants with authenticated identities.
- **Adversary:** a fixed set of at most f corrupted participants can collude.
- **Cryptography:** honest signatures remain unforgeable; messages bind to an instance.
- **Network:** reliable channels, eventual synchrony, and possibly adversarial scheduling.

Target: safety throughout delays; liveness after sufficient network stability and effective honest leadership.

PBFT: four servers running one service

- PBFT means Practical Byzantine Fault Tolerance.
- A client sends a request to a replicated service.
- With four replicas R0, R1, R2, R3, at most one may be Byzantine.
- R0 is the current primary; the other three are backups.

Example request T: transfer 10 units from account X to account Y.

The normal PBFT conversation

Stage	Who sends to whom	Meaning
REQUEST	Client to primary	Please execute T
PRE-PREPARE	Primary to backups	Place T at this log position
PREPARE	Backups to all replicas	I accept this proposal here
COMMIT	Prepared replicas to all replicas	I have enough preparation evidence
REPLY	Executing replicas to client	Here is T's result

The replicas must agree on both the request and its position in the ordered log.

PBFT step 1: the primary proposes

- The client submits request T .
- R_0 sends $\text{PRE-PREPARE}(\text{view } 0, \text{slot } 7, \text{digest}(T))$ with T .
- Each backup checks the request, authentication, view, slot, and digest.
- An honest backup accepts no second conflicting proposal for the same view and slot.

Accepting a proposal is preliminary. The replica has not executed T yet.

PBFT step 2: backups prepare

- R1, R2, and R3 multicast matching PREPARE messages.
- A replica becomes prepared when it has T, the matching PRE-PREPARE, and $2f$ matching PREPARE messages from distinct backups.
- Here $f = 1$, so it needs two backup PREPARE messages.
- A backup may count its own PREPARE.

For R1: R0's proposal + PREPARE from R1 + PREPARE from R2 is enough to become prepared.

PBFT step 3: commit, then execute

- A prepared replica multicasts COMMIT for the same view, slot, and digest.
- To become committed locally, it must be prepared and collect $2f+1$ matching COMMIT messages.
- With $f = 1$, three commits are required, including its own if available.
- It executes only after earlier log slots have executed.

For R1: COMMIT from R0, R1, and R2 is sufficient if R1 is also prepared.

PBFT step 4: the client accepts a result

Replica reply	Result in our example
R0	Transfer completed
R1	Transfer completed
R2	May arrive later
R3	May lie or stay silent

- The client waits for $f+1$ matching replies from distinct replicas.
- For one tolerated fault, two matching replies are sufficient.

Replica commit threshold: three. Client reply threshold: two. They serve different purposes.

A faulty primary and a new view

- R0 may stop proposing or send conflicting proposals for one slot.
- Honest backups reject conflicting local proposals and may time out.
- A new primary gathers a quorum of VIEW-CHANGE evidence.
- Its NEW-VIEW must preserve requests required by that evidence; backups verify it.

Changing leaders does not erase the earlier protocol history.

LSP and PBFT: what changed?

Question	LSP oral messages	PBFT
Main task	One commander's value	Repeated ordered service operations
Mechanism	Recursive relay + majority	Proposal, prepare, commit, view change
Timing	Synchronous rounds	Safety under delay; timed liveness
Authentication	Immediate sender identity	Cryptographic message authentication
Communication	Many recursive message paths	Usually quadratic normal-case traffic

Both need explicit assumptions. PBFT also needs deterministic execution and correct leader-change rules.

A final classroom scenario

- Four validators maintain one ledger. At most one is Byzantine.
- The leader sends different proposals to different honest validators.
- At the same time, some honest messages arrive late.
- Discuss: Which behavior is equivocation? Which is a network delay? What must remain safe?

A useful BFT claim states the fault bound, attacker powers, timing model, and exact guarantees.